BIN381 - Milestone 4

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Executive Summary

The National Treasury must divide a limited development budget between subsidies for household digital access and continued spending on education and usually plans the two separately. This project tests whether that makes sense when their end goal is employment. Does education and home internet access reinforce each other, can one make up for the other or does each work on its own and we refer to these three questions throughout the project as complements, substitutes and independent factors. We used Statistics South Africa's General Household Survey 2024 data and followed the CRISP-DM process from understanding the business to deployment.

After we finished evaluating our models and comparing results we came to the conclusion that education and home internet complements one another when it comes to employment. In a survey-weighted logistic regression for 28742 people in the labour force, the interaction between having matric and having home internet has an odds ratio of 1.29. Among people with matric, home internet goes with about 6 more people employed out of every 100 and among people without matric the difference is only about 2 in 100, which could be chance. When we tested using a random forest model that was never told about the interaction it also found the same pattern. For salaries the interaction was not significant, so income is reported as a side finding.

The selected model meets every success criterion we set, with a test-set AUC of 0.765 against 0.727 for a demographics-only model. It is fully deployed in an R Shiny app that shows group employment rates, province scenarios and checks on new survey data and a Power BI dashboard that gives analysts the descriptive picture with 8 pages of information.

Our Final Recommendation: The National Treasury should plan digital-access and education funding together instead of as competing budget lines, starting where people are educated, but not connected. These results are associations from one 2024 survey, so they should guide pilot programmes, not be read as proof that a subsidy will create a set number of jobs.

1. Business Understanding

The problem: South Africa combines high unemployment with unequal access to the internet and with job seeking now happening online, through job adverts, online applications and staying in contact with employers, not everyone has a fair and equal chance at getting a job (Oyedemi and Choung, 2020). Poor connectivity, high data costs and limited digital skills keep many job seekers out of the process of applying for jobs online. Government programmes such as South Africa Connect aim to close the gap, but access is still uneven (Department of Communications, 2013). Education strongly affects employment and the open question is whether internet access changes how much education pays off. If it does, money spent on education may be under-delivering in places where people cannot get online.

The stakeholder: The National Treasury advises government on how development funding is allocated and each year it must decide how to split a limited budget between digital-access subsidies and education programmes. To do this well, it needs evidence on whether the two policy areas affect each other or can safely be treated separately, which is where we come in.

The business question: Do education and household internet access act as complements, substitutes or independent factors when it comes to employment? International research shows that either pattern can

occur, depending on the country and the group studied (Plavgo, 2023). We therefore tested which one applies to South Africa's working-age population instead of assuming it.

Answer	Meaning	Funding implication
Complements	Each makes the other worth more. People with matric and home internet do better than the two effects added together.	Plan and target both budgets together.
Substitutes	One can make up for the other. Internet helps most where education is low.	Internet access can partly compensate where education is lacking.
Independent	Each has its own effect and neither changes the other.	Fund and evaluate the two separately.

Analytics goal: Model whether a person in the labour force is employed, using education, home internet access and the interaction between the two, while controlling for age, sex, population group, household make-up and location. As a secondary and exploratory question, model the salary of employed people with the same inputs.

Success criteria: The project would succeed if it gave the National Treasury a clear, evidence-based answer to the business question and was honest about the limits of the data.

2. Data Understanding and Preparation

Our data sources come from the eleven datasets supplied Statistics South Africa. D01 to D10 come from one survey that has the same structure as the General Household Survey 2024 (Statistics South Africa, 2025). D11 comes from a separate labour force survey.

The following datasets are what we chose for our analysis:

Dataset	Level	What it gives the project
D01 Population Profile	Person	Age, sex, population group, province, settlement type and survey weights
D02 Learning Profile	Person	Education level and attendance
D03 Economic Participation	Person	Employment status and salary
D08 Connectivity Access	Household	Fixed and mobile internet, computer and smartphones

D04 to D07 and D10 describe housing, services, energy, food security and transport, which are not part of the education, internet and employment pathway. D09 was left out because its only digital column repeats D08 and its income columns partly give away whether a person is employed. D11 could not be used at all, because it shares none of D01's 20940 household IDs. Employment therefore comes from D03 only.

Data quality: No dataset was assumed to be clean. Each one was checked for completeness, duplicates, valid values, consistent labels and whether its keys linked to the other files.

The following contains the main data quality problems and how they were treated:

Problem	Treatment and evidence
Household IDs are 18 digits long and R rounded them on import, which merged the different households (50+ unique households instead of 1000+).	All IDs were read as text. Every dataset then showed the same 20940 distinct households.
25 people in D01 had no household ID.	Each was matched to the same person in D02 and D03 using person number, sex, province and survey weight. No person was lost.

Problem	Treatment and evidence
90 missing and 18 impossible ages in D01, such as -7 and 175.	Replaced with the age recorded for the same person in D03, which had no gaps and agreed with D01 wherever D01 was valid.
120 missing education codes in D02.	Filled from D03, which records the same variable for the same people.
Repeated person keys: 23 in D01, 44 in D02 and 40 in D03.	The D01 cases were caused by the missing IDs. In D02 and D03 they were exact copies, so 22 and 20 rows were removed.
One salary field held almost only the value 1. The other held the codes 888888888 or 999999999 in 74% of rows.	The first field was dropped. Comparing the codes with employment status showed they mean "not applicable" for people without a job and non-response among the employed. Real salaries were kept with a reason recorded for every gap.
Province names in three different casings, which turned 9 provinces into 27 and 15 households with a negative number of smartphones.	Province names were standardised. The smartphone counts were set to missing, keeping the household's other answers.

Analytical grain and integration: Both outcomes belong to individuals, so the analysis uses one row per person aged 15 to 64, which is Statistics South Africa's working-age definition. Each person carries their household's internet and device answers. Relationships were created between D01, D02 and D03 which were joined one-to-one on household ID and person number, which acted as a composite key, and D08 was joined many-to-one on household ID. Every join was set to stop with an error if a key matched more rows than expected or if a row would be dropped. The integrated dataset holds 45413 people in 19780 households, with no rows lost or duplicated.

Engineered features: Twenty-four features were built from fixed rules, so none of them learns from the data and none can leak information from the test set. The most important are:

- **matric_plus,** whether a person has matric or a higher qualification. This gives one clear education measure for the interaction.
- **any_home_internet**, whether the household has fixed or mobile internet. This is the policy lever.
- **Household make-up**, such as household size, the number of children and whether another adult in the household has matric.
- **age_centred** and its square, because employment rises with age and then falls, so a straight line would not fit.
- **settlement_type**, which combines area type and metro status into metro urban, non-metro urban, traditional and farms.
- **log_salary**, the natural log of salary, which stops a few very high earners from dominating the income model.

Modelling data and split: The employment target compares employed and unemployed people in the labour force. People who are not economically active, mainly students and people close to retirement, were left out because the online job-search mechanism does not apply to them. After removing 104 people whose household had no internet answer, 28742 people remained, 18641 employed and 10101 unemployed. Columns that would give away the answer, such as salary, which only employed people have, were removed, with income data holding 1017 employed people with a salary of at least R100.

People in the same household share every household answer, so a normal random split would put near-copies of the same household in both the training and test sets and make the model look better than it is. The split was therefore made by household, with 80% of households for training and 20% for testing

and a fixed seed of 67. This gave 23041 training and 5701 test rows for employment, and no household appears on both sides. The share employed stayed almost the same, at 64.9% in training and 64.7% in testing.

3. Modelling

Employment has a known yes or no outcome, so this is a supervised classification task with all analysis being done in R.

Models Used:

Model	Role	Why it was chosen
Survey-weighted logistic regression	Main explanatory model	The only candidate that estimates the education and internet interaction directly, as an odds ratio with a significance test.
Decision tree	Readable check	Finds interactions without being told about them and produces simple if-then rules.
Random forest	Predictive benchmark	Captures complex patterns the regression cannot, which shows how much accuracy the simpler model gives up.
Demographics-only baseline	Point of comparison	Uses only age, sex, population group, province and settlement type, so it shows what education and internet add.

All employment models use the same 15 predictors, so the comparison is fair. The logistic regression, fitted with the survey package, adds the interaction between matric and home internet as an explicit term (Lumley, 2026). The decision tree and random forest can form interactions on their own by splitting on one variable inside another. For income, a survey-weighted linear regression and a random forest were fitted to log salary using the 8 predictors that the income data contains.

Survey weights: Statistics South Africa sampled some groups more heavily than others and people in the same household are not independent. The regressions therefore use the survey's person weights, so that results describe the South African labour force and not only the sample and they treat households as clusters, so that the significance test is not overconfident. The tree-based models were not weighted as they are used to check the pattern and benchmark accuracy and in those methods a few very heavily weighted people would dominate the splits.

Tuning and validation: All tuning used only the training data, through five-fold cross-validation grouped by household, for the same reason the train and test split was grouped. The decision tree's depth, minimum group size and pruning level were tuned over 36 combinations, and the simplest tree within one standard error of the best was kept, because the tree was chosen for readability. The random forest's main settings were tuned for the highest cross-validated AUC. Since 65% of people are employed, the usual probability cutoff of 0.5 labels almost everyone as employed. Each model's cutoff was therefore tuned on the training folds to give equal weight to employed and unemployed people. For the logistic regression a cutoff of 0.639 raised the share of unemployed people correctly identified from 46% to 68%. The test set was used only once, after all of these choices were fixed.

4. Evaluation

Model performance: Below we tested each model on 5574 people from 3317 households the models had never seen. AUC measures how well a model separates employed from unemployed people, where 0.5 is guessing and 1.0 is perfect, Balanced accuracy is the average share of employed and unemployed people classified correctly, and the Brier score measures how close the predicted probabilities are to what happened, where lower is better.

Results from the Test dataset:

Model	AUC and 95% CI	Balanced accuracy	Brier score	Tests the interaction
Demographics-only baseline	0.727 (0.713-0.741)	0.664	0.194	No
Logistic regression, selected	**0.765 (0.752-0.778)**	**0.693**	**0.183**	**Yes, directly**
Decision tree	0.710 (0.696-0.724)	0.656	0.198	No
Random forest	0.774 (0.761-0.786)	0.699	0.180	Only indirectly

Every model passed the AUC threshold of 0.70, including the baseline, so the threshold alone does not separate them, however the comparison with the baseline does. The logistic regression scores 0.038 higher and a DeLong test run with the pROC package shows this is very unlikely to be chance, with p below 0.0001 (DeLong et al., 1988). Education and internet access therefore add real information beyond who and where a person is. Test results were close to the cross-validated results, so the models are not overfitted and their predicted probabilities matched observed rates well.

The answer to the business question: In the logistic regression, the interaction between matric and home internet has an odds ratio of 1.29, with a 95% confidence interval of 1.06 to 1.58 and p = 0.013 from a Rao-Scott likelihood-ratio test, which adjusts for the survey design. An odds ratio above 1 means home internet goes with a larger employment advantage for people with matric, so the two are complements and because the whole interval lies above 1, a chance result is unlikely. Each person in the training data was given each combination of matric and internet in turn while everything else about them stayed the same and the predictions were averaged.

The expected Employment Rates:

	No home internet	Home internet	Difference and likely range
Below matric	60.9%	62.7%	+1.8 points, from -1.1 to +4.8
Matric or higher	63.5%	69.7%	+6.3 points, from +3.1 to +9.4

Home internet goes with about 6 extra employed people per 100 among people with matric, but only about 2 per 100 among people without matric and that smaller gap could be chance. The difference between the two gaps, about 4 extra people per 100, is the added benefit of having both. Differences were worked out before rounding.

Robustness: Two checks support the finding. The random forest, which was never told about the interaction, found the same pattern in its own predictions, although smaller at about 1.4 points. Removing the computer and smartphone controls barely changed the interaction, which became an odds ratio of 1.28 with p = 0.018. The decision tree never split on education or internet access and is not evidence against the result. About 85% of the labour force has home internet and almost everyone with internet also has a smartphone, so once a tree has split on smartphones, internet adds little.

Income: Both income models beat their baselines, but explain only about 30% of the differences in log salary, with an R^2 of 0.30 for the regression and 0.32 for the random forest. The interaction was not significant for salary, at +3.1% with p = 0.60. Among people who have a job, there is no evidence that

home internet changes the salary advantage of matric, so income is reported as a side finding and is not used for funding decisions.

Final model selection: The random forest scored slightly higher, by 0.009 AUC, but its predictions correlate with the regression model's at 0.93 and give the same result for 89% of test people. The survey-weighted logistic regression was selected, because it is the only model that answers the business question directly, it uses the survey weights so its estimates describe the population, and every result can be traced back to odds ratios a policy analyst can read, while the random forest is kept as a benchmark. The survey-weighted linear regression was selected for income for the same reasons.

5. Shiny App

The selected employment model is deployed as an interactive R Shiny app. Shiny was chosen because it runs the model in the same R environment it was built in and can validate, score and monitor new data. A plumber or vetiver API has no screens for analysts, a static Quarto dashboard cannot check uploaded data and R visuals in Power BI are static images. The income model is not deployed as a predictive tool, because its interaction was not significant, but its result is summarised in the app. Each of the app's five pages starts with a short guide on how to read it, then gives the answer in plain words, followed by the supporting numbers.

- **Overview** gives the answer to Treasury's question, what it means for funding and how far the results can be trusted.
- **Education × internet** shows expected employment rates for the four education and internet groups, filtered by province, area type, age band and gender.
- **Province scenarios** shows what the model implies if a chosen share of people had home internet. It is labelled clearly as a "what if" and not a forecast.
- **Check new data** lets an analyst upload a CSV file of new survey data, which the app validates, scores and checks.
- **About the model** explains how the model was chosen, how accurate it is for each group, its limits and when to retrain it.

Protecting the survey data: The saved model object contains a full copy of the training data, so the app uses a slim version that holds only the 30 coefficients, their covariance matrix, the category definitions and group totals. It predicts identically and contains no survey records, so the app could be published online without exposing anyone's data.

Checking new data: Each uploaded file is checked for the 15 required columns, valid values and categories, and agreement between related columns before it is scored. If the file includes the real employment outcome, the app compares accuracy, calibration, changes in the mix of people and the accuracy gap between population groups with the test-set results and marks each check as OK, Investigate or Action needed. The app only shows results for groups and whole files, never a prediction for an individual person.

6. Power BI Decision-Support Dashboard

The Power BI dashboard gives the National Treasury analysts a descriptive view of the same story, built on the final analytical dataset of 45413 working age people and the model results exported from R. Instead of showing every variable, each page answers one decision question.

Page	Decision question	What it shows
Overview	What is the overall picture?	Labour force, employment rate, matric share, home internet share and the extra boost from having both
Education x Internet	Do education and internet work best together?	Employment rate for each education and internet group, the internet gap with and without matric, and a province breakdown
People without internet	Who is not connected?	Home, mobile and fixed internet and computer access by province and area type
Does Education Pay?	How much does education add on its own?	Employment rate by education level and sex, the matric advantage and matric share by province
Where to Act	Where would more internet access help most?	A province map with today's employment and internet rates and the modelled gain from the province scenarios
Model and Trust	How far can the results be trusted?	AUC of all the models and how well the model identifies employed and unemployed people in each population group

The dashboard adds two points that shape the recommendations. Home internet in South Africa mostly means mobile data, since only about 1 in 5 people has a fixed line and the Northern Cape and Eastern Cape have the lowest access. The Model and Trust page also places group-level accuracy next to the headline results, so a user cannot mistake one overall accuracy figure for equal reliability across groups.

7. Findings and Recommendations

Finding	Evidence	Recommendation
1. Education and home internet are complements for employment.	Odds ratio 1.29, 95% CI 1.06 to 1.58, p = 0.013. Confirmed by the random forest and without device controls.	Plan digital-access and education funding together instead of as competing budget lines.
2. Home internet matters most for people with matric.	+6.3 employed per 100 with matric, compared with +1.8 per 100 without.	Start connectivity spending where many people have matric but little home internet.
3. Internet access on its own shows a weak link with employment.	The +1.8 gap for people without matric could be chance.	Do not present connectivity as a stand-alone job programme. Pair it with education support.

Finding	Evidence	Recommendation
4. Access is lowest in some provinces and in traditional areas.	The Northern Cape and Eastern Cape have the lowest home internet, and fixed lines are rare.	Use the province scenarios in the Shiny app and dashboard to choose pilot provinces.
5. There is no income interaction.	+3.1% salary difference, p = 0.60.	Do not use these results for income-support decisions.
6. Model accuracy differs between groups.	One cut-off identifies 71% of unemployed Black African respondents but only 4% of unemployed White respondents.	Use the model for group-level planning only, and always show group accuracy next to headline figures.

These recommendations rest on associations from one 2024 survey. They show that education, internet access and employment go together, not that providing internet will cause a set number of people to find work. The direction could even run the other way, because having a job makes it easier to pay for data and a smartphone. The results should therefore guide pilot programmes that can measure the effect directly.

8. Limitations, Assumptions and Future Work

8.1. Limitations

- Association, not cause**:** A single cross-section cannot show what would happen if a person gained matric or internet access.
- One source for employment**:** D11 could not be linked, so the employment outcome comes only from D03.
- Household, not personal, internet**:** Internet access is a household answer, so the results describe living in a connected household, not personal use.
- Household wealth is not controlled**:** D09 was excluded, so wealth may explain part of the internet effect.
- Uneven accuracy between groups**:** The AUC is similar across groups, but one cut-off meets very different employment rates. Where more than 85% of a group is employed, the model rarely predicts unemployment, and in traditional areas it misses more than half of the employed people.
- Income explains little**:** The income models explain about 30% of salary differences, partly because occupation, industry and hours worked are not available.
- Smaller issues**:** The DeLong tests and AUC intervals treat people as independent, although household members are not, so uncertainty is slightly understated. 497 people with an unknown education level are left out of models that use matric.

8.2. Assumptions

We assumed that D01 to D10 describe the same households and people, which the joins confirmed, and that the survey weights supplied by Statistics South Africa correctly represent the 2024 population. No codebook was supplied, so we assumed the value labels in the data dictionary match Statistics South Africa's definitions. We also assumed that salaries are monthly, based on their size, because the pay period is not documented, and that the 2024 survey still reflects conditions when the tools are used.

8.3. Future work

The most useful next steps are to add an asset-based wealth measure from D09 as a robustness check, re-test the interaction on each new survey wave, measure personal rather than household internet use, and use quantile regression to see whether internet affects low and high earners differently. A pilot programme that adds connectivity in selected areas would be the strongest way to test whether the relationship is causal.

9. Ethics and Responsible AI

Phase	Issue	What we did
Business and data understanding	Privacy of survey microdata	Used the de-identified data for academic purposes only and flagged the risk of identifying people in very small groups.
Data preparation	Re-identification	Removed metro_code, which splits provinces into 17 municipalities. Household IDs are used only for joining and splitting, never as predictors.
Modelling	Sensitive attributes and representativeness	Kept population group and sex as controls, because dropping them hides inequality without removing it. Survey weights make results describe the whole labour force.
Evaluation	Unequal accuracy between groups	Measured performance for each population group, sex and settlement type, and reported the gaps openly.
Deployment	Misuse for decisions about individuals	Group results only, no population-group filter, groups under 50 people hidden, and a monitoring check that flags a widening fairness gap.

Fairness findings**:** The model ranks people about equally well within each group, with an AUC between 0.70 and 0.83. The problem is the single cut-off. It correctly identifies 71% of unemployed Black African respondents, 41% of unemployed Coloured respondents, 20% of unemployed Indian/Asian respondents and only 4% of unemployed White respondents, because more than 85% of the last two groups are employed. It also misses more employed women than men. These gaps most likely reflect real inequality in the labour market, with some sampling noise in the smallest groups, so the model is used only for group-level planning.

Responsible AI use**:** The team used Claude, an AI assistant, to help debug and validate R code, build the Shiny app and its synthetic test file, check how results were interpreted, and improve the wording and structure of the documentation, including this report and the presentation. All AI-assisted code was rerun and checked against the R output, and all text was read and corrected by the team. The full usage log is in Appendix B.

10. Project Review

10.1. Team Reflections

What worked**:** Organising the code by CRISP-DM stage, with one shared preparation function for every notebook and the app, meant all data was always prepared in the same way, and a fixed seed of 67 made every result reproducible.

What surprised us**:** Several early conclusions were wrong. The 74% of "unusable" salaries were mostly people without a job, the household IDs were silently rounded on import, and the decision tree ignored education and internet even though the regression found a clear interaction.

What changed across milestones**:** Milestone 1 planned to show the analysis through R visuals in Power BI. By Milestone 3 deployment moved to Shiny, because only Shiny could validate and score new data, and Power BI became the descriptive dashboard. The research also narrowed to the employment question.

What we learnt and would improve**:** We learnt why survey weights and household-grouped splits matter, and that accuracy alone misleads when one class is much larger. Shiny was new to us. Next time we would add a wealth control from the start, check group-level accuracy earlier and build the dashboard alongside the modelling.

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Appendix: Repository, Reproduction and Monitoring

A. Repository Structure

```
BIN381/
├── BIN381.Rproj                    RStudio project (open this first)
├── README.md
├── Scripts/
│   ├── _quarto.yml
│   ├── 1) EDA.qmd
│   ├── 2.1) Data Preprocessing.qmd
│   ├── 2.2) Train-Test Split.qmd
│   ├── 3.1) Employment Model.qmd
│   ├── 3.2) Income Model.qmd
│   ├── 3.3) Model Evaluation.qmd
│   ├── 3.4) Deployment Preparation.qmd
│   └── R/
│       └── model_preparation.R
│
├── Shiny-App/
├── Models/
├── Datasets/
│   ├── D01_…csv – D11_…csv
│   ├── Cleaned/
│   └── Analytical/
│
├── Documentation/
│   ├── Milestone Documentation/
│   │   ├── Milestone 1/
│   │   ├── Milestone 2/
│   │   ├── Milestone 3/
│   │   └── Project Outline/
│   └── Rendered Documentation/
│
└── Power Bi/
    ├── Final Power Bi.pbix
    ├── BIN381_Theme.json
    ├── Initial Power Bi.pbix
    └── Dashboard_*_Theme.json
```

B. To Reproduce

a. Requirements

You need R and RStudio with Quarto. The eleven raw Statistics South Africa files (D01 to D11, .csv) must be in Datasets/. Install the packages once:

install.packages(c("tidyverse", "janitor", "skimr", "visdat", "survey", "rpart", "rpart.plot","ranger", "pROC", "caret", "shiny", "bslib"))

b. Setup

Open BIN381.Rproj in RStudio so the working directory is the project root, which the Shiny app needs. The notebooks use paths relative to the Scripts folder (for example ../Datasets/), which is the folder they run in when rendered or run from RStudio.

c. Run order

Run the notebooks in this order. Each one reads files written by earlier steps, so a notebook can be rerun on its own once its inputs exist. 1) EDA.qmd is the Milestone 1 exploration and, because it writes no files it is not needed to reproduce the results.

Step	Notebook	Reads	Writes
1	2.1) Data Preprocessing.qmd	Raw files in Datasets/	Four cleaned files in Datasets/Cleaned/; analytical_dataset.csv, model_data.csv and income_data.csv in Datasets/Analytical/
2	2.2) Train-Test Split.qmd	model_data.csv, income_data.csv	model_train.csv, model_test.csv, income_train.csv, income_test.csv in Datasets/Analytical/
3	3.1) Employment Model.qmd	model_train.csv	employment_logistic.rds, employment_tree.rds, employment_forest.rds in Models/
4	3.2) Income Model.qmd	income_train.csv	income_linear.rds, income_forest.rds in Models/
5	3.3) Model Evaluation.qmd	Train and test files; employment_logistic.rds, income_linear.rds	Tuned and baseline models and employment_thresholds.csv in Models/
6	3.4) Deployment Preparation.qmd	model_data.csv, train and test files; models and thresholds from steps 3 to 5	Shiny-App/app_data.rds

Every random step uses seed 67, so reruns produce identical files and models. Rendering a notebook (the Render button in RStudio, or quarto render in the Scripts folder) saves the output in Documentation/Rendered Documentation/; Scripts/_quarto.yml sets this.

d. Running the Shiny app

From the project root, run shiny::runApp("Shiny-App"). The app loads only app_data.rds, so it contains no survey microdata. To try the Check new data page, upload Shiny-App/dummy_employment_test_data.csv or Datasets/Analytical/model_test.csv.

The app is also hosted at https://henri123.shinyapps.io/bin381-digital-access/ .

e. Check that it ran correctly

After step	Expected result
1	45,413 people in 19,780 households; 28,742 in the employment population (18,641 employed, 10,101 unemployed); 18,017 in the income population
2	Employment 23,041 training and 5,701 test rows; income 14,371 training and 3,646 test rows
5	Test AUC 0.765 (logistic regression, selected), 0.727 (baseline), 0.710 (decision tree) and 0.774 (random forest); threshold 0.639; interaction odds ratio 1.29 (95% CI 1.06 to 1.58), p = 0.013
6	app_data.rds created in Shiny-App/

C. Monitoring

Check	**Test-set value**	**OK**	**Investigate**	**Action needed**
AUC	0.765	0.715 or higher	0.70 to 0.715	Below 0.70
Balanced accuracy	0.693	0.65 or higher	0.60 to 0.65	Below 0.60
Brier score	0.183	0.20 or lower	0.20 to 0.22	Above 0.22
Calibration gap	0.005	Within ±0.03	±0.03 to ±0.05	Beyond ±0.05
Drift in the mix of people	About 0	Below 0.10	0.10 to 0.25	Above 0.25
Widening of the AUC gap between population groups	0.133	Up to +0.05	+0.05 to +0.10	Above +0.10